

Unit 1: Definition and Components of MIS

Introduction to Information Systems :

An Information System (IS) is an organized combination of people, technology, data, processes, and networks that collects, processes, stores, and distributes information. Information systems help individuals and organizations perform daily activities, solve problems, make decisions, and achieve their goals.

Main Components of an Information System

1. **Hardware** – Physical equipment such as computers, servers, printers, and mobile devices.
2. **Software** – Programs and applications that instruct hardware what to do.
3. **Data** – Raw facts that are processed into meaningful information.
4. **People** – Users and IT professionals who develop, operate, and use the system.
5. **Processes/Procedures** – Rules and methods that determine how the system is used.
6. **Networks** – Communication technologies that connect computers and allow information to be shared.

Data vs. Information

Data consists of raw facts, such as individual sales transactions. Information is data that has been processed and

organized so that it has meaning. For example, calculating the total monthly sales from individual transactions converts data into useful information.

Types of Information Systems

Common types include Transaction Processing Systems (TPS) for routine transactions, Management Information Systems (MIS) for reports and management activities, Decision Support Systems (DSS) for decision-making, and Enterprise Systems such as ERP and CRM systems that integrate activities across an organization.

Importance of Information Systems

Information systems can improve efficiency, communication, decision-making, data management, customer service, and competitive advantage. For example, a supermarket information system can record purchases, update inventory automatically, process payments, and provide managers with sales reports.

In simple terms: an information system takes data as input, processes and stores it, and produces useful information as output to support people and organizations.

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Transaction Processing System(TPS) :

A Transaction Processing System (TPS) is an information system that is used to record, process, and manage routine day-to-day transactions of an organization.

Definition

A Transaction Processing System is a computer-based system that collects and processes data generated from business transactions and produces useful records and information.

Simple example:

When you purchase a product from Amazon, the system records your order, payment, product details, and updates the inventory. This is transaction processing.

Examples of TPS

- **Sales system** – Records product sales
- **Banking system** – Processes deposits, withdrawals, and fund transfers
- **Payment system** – Processes card/online payments
- **College system** – Records student admission and fee payments
- **Reservation system** – Handles flight/hotel bookings
- **Payroll system** – Calculates employee salaries

Basic Working of TPS

Input → Processing → Storage → Output

1. **Input:** Transaction data is entered.
2. **Processing:** The system validates and processes the data.
3. **Storage:** Transaction details are stored in a database.
4. **Output:** Receipts, reports, confirmations, etc. are generated.

Characteristics of TPS

1. **High volume:** Handles a large number of transactions.
2. **Fast processing:** Transactions are processed quickly.
3. **Accuracy:** Reduces errors in transaction processing.
4. **Reliability:** Must work continuously and correctly.
5. **Data storage:** Maintains transaction records.
6. **Security:** Protects sensitive transaction data.
7. **Real-time or batch processing:** Transactions can be processed immediately or in groups.

Types of Transaction Processing

1. Batch Processing

Transactions are collected and processed together at a particular time.

Example: Monthly salary processing.

2. Real-Time Processing

Transactions are processed immediately after they occur.

Example: ATM withdrawal or online payment.

Importance of TPS

TPS provides the **basic data required by other information systems**, such as MIS and DSS. It helps organizations maintain accurate records and perform their daily operations efficiently.

In short:

TPS is mainly concerned with the routine, day-to-day transactions of an organization.

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Management Information System(MIS) :

A Management Information System (MIS) is an information system that collects, processes, stores, and provides information to managers to help them in planning, controlling, monitoring, and decision-making.

Definition

MIS is a computer-based system that provides managers with timely, accurate, and relevant information in the form of reports for effective management.

Simple Example

Suppose a college has an MIS. It can collect information about:

- Student admissions
- Attendance
- Examination results
- Fees
- Faculty workload

The system can generate a report such as “**Department-wise student attendance**”. The Head of Department can use this report to identify students with low attendance and take appropriate action.

Components of MIS

1. **People** – Managers, employees, and system users.
2. **Hardware** – Computers, servers, printers, etc.
3. **Software** – Applications used to process information.
4. **Data** – Student, employee, sales, financial, and other organizational data.
5. **Procedures** – Rules and methods for using the system.
6. **Network** – Internet and communication infrastructure.

Functions of MIS

1. Data Collection

Collects data from various sources, including TPS.

2. Data Processing

Converts raw data into meaningful information.

3. Data Storage

Stores organizational data in databases.

4. Report Generation

Produces regular and summarized reports for managers.

5. Decision Support

Provides information that helps managers make better decisions.

Characteristics of MIS

- Provides **accurate and relevant information**
- Provides information at the **right time**
- Mainly supports **middle-level management**
- Generates **routine and summary reports**
- Uses data from **Transaction Processing Systems (TPS)**
- Helps in **planning, monitoring, and controlling**
- Improves managerial efficiency

Types of MIS Reports

Report	Meaning	Example
Periodic Report	Generated regularly	Monthly sales report
Exception Report	Shows unusual situations	Students with attendance below 75%
Summary Report	Provides summarized information	Total sales by department
Ad-hoc Report	Created when specifically required	Sales of a particular product

TPS vs MIS

TPS

Handles daily transactions

Used mainly at operational level

Processes detailed data

Example: Sales transaction

Provides data to MIS

MIS

Provides information for management

Used mainly by middle-level managers

Uses summarized data

Example: Monthly sales report

Uses data from TPS

In Simple Words

TPS records what is happening, while MIS tells managers what is happening and helps them monitor the organization.

Example:

TPS: Records 500 product sales today.

MIS: Generates a report showing total sales, department-wise sales, and monthly sales trends.

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Decision Support System(DSS) and Executive Support System(ESS) :

Both DSS and ESS are information systems used to support management decision-making, but they are designed for different levels and types of decisions.

1. Decision Support System (DSS)

A Decision Support System (DSS) is a computer-based information system that helps managers analyze data and make decisions, especially when the decision is semi-structured or complex.

Simple Definition

DSS provides data, analytical tools, and models to help managers make better decisions.

Example

A university wants to decide how many new BCA sections should be opened next year.

A DSS can analyze:

- Previous admissions
- Student demand
- Faculty availability
- Classroom capacity
- Previous-year trends

The manager can then compare different scenarios and make a decision.

Features of DSS

1. Helps in decision-making
2. Supports semi-structured and unstructured decisions
3. Uses data from databases and other systems
4. Provides what-if analysis
5. Can analyze trends and forecasts
6. Allows users to interact with the system
7. Helps managers compare different alternatives

Common Applications

- Sales forecasting
- Financial planning
- Inventory management
- Loan approval
- Student performance analysis
- Production planning
- Resource allocation

Example of What-if Analysis

A manager can ask:

"What will happen to profit if the price is reduced by 10%?"

The DSS can analyze the situation and show the possible result.

2. Executive Support System (ESS)

An Executive Support System (ESS), also called an Executive Information System (EIS), is designed to provide senior executives and top-level managers with summarized and easily understandable information for strategic decision-making.

Simple Definition

ESS provides top-level executives with important summarized information about the organization to support strategic planning and decision-making.

Example

The CEO of a university may want to see:

- Total student admissions
- Revenue
- Overall expenses
- Placement percentage
- Department performance
- Student growth
- Competitor information

An ESS can display this information through dashboards, graphs, charts, and KPIs.

Features of ESS

1. Designed for top-level management
2. Provides highly summarized information
3. Supports strategic decisions
4. Uses internal and external information
5. Provides dashboards and visual reports
6. Shows Key Performance Indicators (KPIs)
7. Helps identify trends and opportunities
8. Easy-to-use interface

DSS vs ESS

Feature	DSS	ESS
Full Form	Decision Support System	Executive Support System
Main Users	Middle-level managers	Top-level executives
Decision Level	Tactical/managerial	Strategic
Information	Detailed and analytical	Highly summarized
Decision Type	Semi-structured/complex	Strategic and largely unstructured
Main Purpose	Analyze alternatives	Monitor overall organization

Feature	DSS	ESS
Examples	Sales forecasting, resource planning	CEO dashboard, organizational performance
Data	Mostly internal + analytical data	Internal + external strategic data
Output	Models, analysis, what- if results	Dashboards, graphs, KPIs

Easy Way to Remember

TPS → Records transactions

MIS → Provides regular reports

DSS → Helps analyze and make decisions

ESS → Helps top executives make strategic decisions

Example:

A sales employee enters a sale → **TPS**

Manager gets monthly sales report → **MIS**

Manager analyzes whether to increase production → **DSS**

CEO reviews overall company growth and decides future expansion → **ESS**.

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Role of MIS in Business Organizations :

A Management Information System (MIS) plays an important role in business organizations by providing accurate, timely, and relevant information to managers. It helps them plan, control, coordinate, and make better decisions.

Major Roles of MIS

1. Supports Decision-Making

MIS provides useful information to managers so they can make informed decisions.

Example: A sales manager uses sales reports to decide which products should be promoted.

2. Planning

MIS helps managers prepare future plans by providing historical data, current performance, and trends.

Example: Using previous sales data to prepare next year's sales targets.

3. Controlling Business Activities

Managers can compare actual performance with planned performance and identify problems.

Example: Comparing actual expenses with the monthly budget.

4. Improves Operational Efficiency

MIS reduces manual work and provides information quickly, helping employees complete tasks more efficiently.

5. Improves Communication

MIS makes information available to different departments and helps improve communication between managers and employees.

6. Helps in Resource Management

It helps organizations manage resources such as employees, money, materials, machines, and time.

7. Performance Monitoring

MIS generates reports and performance indicators that allow managers to monitor employees, departments, products, and overall business performance.

8. Forecasting

MIS can help identify trends and predict future conditions.

Example: Forecasting future product demand based on previous sales.

9. Problem Identification

MIS can generate exception reports that highlight unusual situations or areas requiring management attention.

10. Provides Competitive Advantage

Organizations can use information to respond faster to customers, reduce costs, improve products, and compete more effectively.

Example

Consider an e-commerce company:

Customer orders → TPS → Database → MIS → Management Reports → Managerial Decision

MIS may provide a report showing:

- **Total sales**
- **Best-selling products**
- **Customer orders**
- **Inventory levels**
- **Regional sales**
- **Monthly profit**

Management can then use this information to decide which products to stock, where to increase marketing, and how to improve sales.

In Simple Words

MIS converts business data into useful information that helps managers plan, control, monitor, and make better decisions.

Key role:

Data → Information → Analysis → Management Decision → Better Business Performance.

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System Development Life Cycle(SDLC) :

System Development Life Cycle (SDLC) is a systematic process used to develop, implement, maintain, and improve an information system or software.

Definition

SDLC is a series of stages followed by an organization to develop a high-quality information system in a planned and organized manner.

The main purpose of SDLC is to ensure that the system meets user requirements, is developed within budget and time, and works correctly.

Phases of SDLC

1. Planning

The organization identifies the problem, objectives, scope, cost, and resources required for the new system.

Example: A college decides to develop an online student management system.

2. System Analysis

The requirements of users are collected and analyzed.

Questions include:

- What does the user need?
- What problems exist in the current system?

- What functions should the new system provide?

3. System Design

The structure and working of the system are designed.

It includes:

- Database design
- User interface
- System architecture
- Input and output design
- Security design

4. Development / Coding

Programmers convert the system design into actual software/code using programming languages and technologies.

5. Testing

The developed system is tested to identify and correct errors, bugs, and security issues.

Common testing includes:

- Unit testing
- Integration testing
- System testing
- User acceptance testing

6. Implementation

The completed system is installed and made available to users.

Implementation methods can include:

- **Direct:** Old system is immediately replaced.
- **Parallel:** Old and new systems operate together.
- **Pilot:** New system is introduced to a small part of the organization first.
- **Phased:** System is introduced in stages.

7. Maintenance

After implementation, the system is continuously monitored and updated.

Maintenance may include:

- Fixing errors
- Improving performance
- Adding new features
- Updating security
- Adapting to new requirements

SDLC Flow

Planning → Analysis → Design → Development → Testing → Implementation → Maintenance

Advantages of SDLC

- Provides a structured development process
- Clearly identifies user requirements
- Helps control time and cost
- Improves software quality
- Reduces development risks
- Makes testing and maintenance easier
- Improves communication between developers and users

Simple Example

For a College Management System:

Planning: Decide to computerize student records



Analysis: Identify requirements of students, teachers, and administrators



Design: Design database and screens



Development: Write the program



Testing: Check admission, attendance, fees, and result modules



Implementation: Deploy the system in the college



Maintenance: Fix bugs and add new features

In short: SDLC provides a step-by-step approach for developing a reliable and useful information system.

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Information System Resources :

In Management Information System (MIS), information systems use different resources to collect, process, store, and distribute information for supporting business operations and decision-making.

There are 5 major resources of an Information System:

1. People Resources

People are the most important resource of an information system. They develop, operate, maintain, and use the system.

Examples:

- System analysts
- Programmers
- Database administrators
- Managers
- Employees
- End users

2. Hardware Resources

Hardware refers to the physical devices used for input, processing, storage, and output of information.

Examples:

- Computers and laptops

- Servers
- Keyboard and mouse
- Printers
- Hard disks/SSDs
- Network devices

3. Software Resources

Software consists of programs and instructions that tell the computer how to perform tasks.

Types:

- **System software:** Windows, Linux, Android
- **Application software:** MS Excel, Tally, ERP, payroll systems

4. Data Resources

Data is the raw material that an information system processes to produce useful information.

Examples:

- Customer details
- Employee records
- Sales transactions
- Product information
- Student marks

Data may be stored in databases, data warehouses, or files.

5. Network/Communication Resources

These resources allow computers and people to communicate and share data.

Examples:

- Internet
- Wi-Fi
- LAN
- Routers
- Switches
- Communication networks

Simple Example

Consider a college MIS:

Resource Example

People Teachers, students, administrators

Hardware Computers, servers, printers

Software Student Management System

Data Student attendance, marks, fees

Network Internet, LAN, Wi-Fi

In short:

**People + Hardware + Software + Data + Network =
Information System Resources**

These resources work together to provide accurate, timely, and useful information, to managers for planning, controlling, and decision-making.

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Ethical and Social Issues in Information System :

Information systems provide many benefits to organizations, but they also create ethical and social problems related to privacy, security, employment, and the use of information.

1. Privacy

Privacy means an individual's right to control how their personal information is collected and used.

Examples:

- Collection of customer personal data
- Tracking users on websites
- Sharing personal information without permission

Issue: Organizations must protect personal information and use it responsibly.

2. Security

Information systems can be affected by hacking, viruses, malware, data theft, and unauthorized access.

Examples:

- Hacking of company databases
- Password theft
- Ransomware attacks

Issue: Organizations need proper security measures to protect data and systems.

3. Accuracy of Information

Information stored in an information system should be correct, complete, and up to date.

Incorrect information can lead to wrong decisions.

Example: If a student's marks are entered incorrectly, the system may generate an incorrect result.

4. Intellectual Property Rights

Software, databases, digital content, music, images, and other digital resources may be protected by copyright, patents, or licenses.

Ethical issue: Using or copying software and digital content without permission is unethical and may be illegal.

Example: Using pirated software in an organization.

5. Employment Issues

Information technology and automation can change the nature of jobs.

Examples:

- Automated systems replacing repetitive tasks
- Requirement for employees to learn new technologies

- Reduction in some traditional jobs

At the same time, information systems also create new IT-related jobs.

6. Digital Divide

The digital divide refers to the gap between people who have access to modern technology and those who do not.

This gap may exist because of differences in:

- Income
- Education
- Location
- Internet availability
- Technical knowledge

Example: Students without reliable internet access may face difficulties in online education.

7. Ethical Use of Information

Information should be used honestly and responsibly.

Organizations and individuals should not:

- Manipulate information
- Misuse confidential data
- Spread false information
- Access information without authorization

8. Surveillance and Monitoring

Organizations may use information systems to monitor employees, customers, websites, emails, or online activities.

Issue: Excessive monitoring can violate an individual's privacy.

9. Health and Social Problems

Excessive use of computers and digital devices can cause:

- Eye strain
- Poor posture
- Stress
- Reduced physical activity
- Internet or social-media addiction

Information systems can also affect social interaction and work-life balance.

10. Responsibility and Accountability

When an information system causes harm or produces incorrect results, it is important to determine who is responsible.

For example, if an automated system makes an incorrect decision, responsibility may involve:

- System developers

- Organization
 - System operators
 - Management
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Ethical and Social Issues — Short Summary

Issue	Meaning
Privacy	Protection of personal information
Security	Protection from unauthorized access and attacks
Accuracy	Ensuring information is correct
Intellectual Property	Protecting software and digital content
Employment	Impact of technology on jobs
Digital Divide	Unequal access to technology
Ethical Use	Responsible use of information
Surveillance	Monitoring people through technology
Health & Social Issues	Effects of excessive technology use
Accountability	Determining responsibility for system outcomes

In conclusion: Ethical and social issues are important in MIS because information systems affect individual privacy,

business operations, employment, society, and decision-making. Organizations should use technology in a legal, secure, fair, and socially responsible manner.